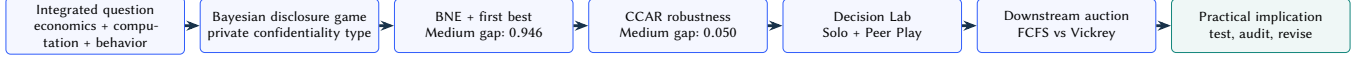


# Share or Hide? Strategic Disclosure in Congested Low-Altitude Drone Traffic

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**Figure 1: Integrated evidence chain.** The three questions meet in one auditable sequence: a Bayesian disclosure model, exhaustive equilibrium and welfare tests, the CCAR mechanism test, an exploratory decision environment, a bounded downstream allocation comparison, and a practical revision gate.

**Submission to:** COMSCI/ECON 206 — Computational Microeconomics.

**Course session:** Autumn 2026 Session 1. **Team:** FP10. **Symposium:** Session B.

**Course instructor:** Professor Luyao Zhang. **Assignment:** PS2 collaborative research proposal.

## 1 Three connected research questions and verified gap

Commercial drone-delivery operators sharing capacity-constrained urban low-altitude corridors can avoid delay by sharing flight intent, yet route and timing detail may reveal demand to competitors. Operators, the traffic-management authority, customers, nearby/public stakeholders, and emergency/public-safety users are affected. Closest verified work designs efficient/fair traffic-flow protocols, limited-information traffic management, privacy-preserving coordination, and vertiport auctions [1, 2, 4, 6]; oligopoly research explains incentives to share private information [5]. The narrow surviving gap is endogenous *granularity* of voluntary, non-safety disclosure when commercial-confidentiality cost is private and coordination value changes with congestion—not a claim that privacy or airspace auctions are new.

**Q1—Economics:** Under incomplete information about commercial-confidentiality costs, how does congestion change equilibrium disclosure, and when does equilibrium disclosure differ from the full-information first best?

**Q2—Computation:** How do equilibrium disclosure, welfare loss, and candidate-mechanism performance vary across congestion, confidentiality cost, and access-rule parameters?

**Q3—Behavior:** Do participants depart from the Bayesian benchmark because of privacy salience, reciprocity, fairness, competitive concerns, or misunderstanding of strategic uncertainty?

**Integrated question:** Which disclosure-and-access design remains welfare-improving, behaviorally intelligible, and institutionally accountable across private costs and congestion? Figure 1 connects the questions: incentives define the model, computation tests it, behavior probes its assumptions, and allocation plus a revision gate bound the practical claim.

## 2 Answer to Q1: incentives and strategic model

Players  $i \in \{A, B\}$  simultaneously choose  $a_i \in \{M, P, F\}$  (Minimum, Partial, Full voluntary commercial disclosure); mandatory safety information is outside the action set. Nature independently draws private  $c_i \in \{1, 1.6\}$  with prior  $(0.5, 0.5)$ ; public congestion  $d \in \{0.3, 0.6, 1\}$  is known before choice. A complete pure strategy maps each own type to an action. With  $x = (0, 1, 2)$ ,  $k = (0, 1, 2.5)$ ,  $B = 7.5$ , and  $\lambda = 0.5$ ,

$$u_i(a_i, a_j; c_i, d) = dB[1 - e^{-\lambda(x_i + x_j)}] - c_i k(a_i).$$

Simultaneous choice plus an unobserved rival confidentiality type makes pure-strategy Bayesian Nash equilibrium (BNE) the appropriate benchmark [3]. Exhaustive interim best-response enumeration gives the unique type-contingent BNE  $(L, H)$ : Low  $(M, M)$ , Medium  $(P, M)$ , High  $(P, P)$ .

**Game theory** predicts  $M/P/F$  disclosure by type. **Social choice** compares  $W = u_A + u_B$  with the realized-type full-information first best, exposing coordination/privacy and efficiency/fairness trade-offs: BNE/first-best/gap is  $0/0.639/0.639$ ,  $2.193/3.139/0.946$ , and  $6.882/7.385/0.503$ . **Mechanism design** changes enhanced non-safety information access through CCAR and tests incentive/welfare robustness: at Medium,  $\alpha_M = 0.7$  changes BNE to  $(P, P)$  and underlying welfare to 3.089 (gap 0.050), without restricting safety information.

## 3 Answer to Q2: computation, auction comparison, and artifacts

**Language-independent pseudocode:** for each congestion state, enumerate type-contingent strategies; compute interim utility over rival types; retain mutual best responses as pure BNE; for each realized type pair enumerate action profiles and maximize full-information welfare; compare expected BNE and first-best welfare; repeat under CCAR and parameter sweeps. Inputs are the stated primitives; baseline is unrestricted enhanced access, comparison is CCAR, and metrics are equilibrium actions, welfare/gap, multiplicity, and failure diagnostics. Fresh runs yield the results above; a 760-cell sweep finds 401 no-effect, 334 robust-improvement, 18 excessive-disclosure, 3 increased-gap, 66 multiple-BNE, and 0 no-pure-BNE cells (overlapping diagnostics, not probabilities).

The bounded application allocates one residual commercial priority slot (emergency/public-safety users excluded) among operators with private values and sealed reports. FCFS allocates to the first eligible request and stops; a single-unit second-price auction allocates to the highest bid, charges the second bid, breaks ties by ID, and stops. Under the independent-private-values rational benchmark, truthful bidding is weakly dominant [7]. For values (8, 5, 3, 2) and arrival  $B, C, A, D$ , FCFS gives  $B$  efficiency 0.625; truthful second price gives  $A$ , payment 5, utility 3, efficiency 1. In 10,000 seed-206 rounds, mean efficiency is 0.622308 versus 1.000000 and auction revenue is 5.99207. Ability to pay, urgency, interdependent values, and legitimacy bound the analogy.

**Artifacts:** GitHub at current reproducibility checkpoint ffa4836208a3f647181771646c5639286c0726e6; Colab 01; Hugging Face; validated editable poster PS2-FP10-Yiqiao-A0-Poster.pptx. The two notebook paths, both hosted Run-all confirmations, and all versions appear in Appendix A.

#### 4 Answer to Q3: behavior and interdisciplinary integration

The rational benchmark is type-contingent BNE disclosure. Candidate mechanisms are *privacy salience*, *reciprocity/fairness*, and *competitive concern*; the competing explanation is misunderstanding incomplete-information incentives or the payoff structure. The Hugging Face Solo and same-device Peer Play environment is exploratory evidence infrastructure: choice precedes benchmark reveal, then CCAR and reflection probe responses. No representative behavioral sample exists. A planned discriminating test compares initial choice, post-benchmark reflection, CCAR second choice, and a trusted-partner counterfactual: stable deviations after comprehension checks that move with framing/partner trust favor preference or framing accounts; convergence after explanation favors misunderstanding. Evidence would qualify the benchmark or revise the interface, not establish population effects.

#### 5 Advanced development, real-world impact, and roadmap

For commercial drone-delivery operators sharing capacity-constrained urban low-altitude corridors, the decision is how much non-safety flight-intent information to share and how to allocate residual commercial priority scarcity. Potential benefit is coordination and avoided delay; risks are commercial exposure, unequal access, over-disclosure, and misspecified incentives. Before adoption, the authority, operators, customers, nearby/public stakeholders, and emergency/public-safety users need pilot calibration of latency and confidentiality value, safety review, and distributional analysis. Correction requires auditable rules, a safety-information floor, appeal/review, and parameter revision or suspension when welfare/access diagnostics fail. Academia requires equilibrium, identification, robustness, and reproducibility; industry requires feasibility, reliability, confidentiality, and latency; public institutions require safety, access, legitimacy, auditability, and correction. Next: preregister the behavioral discrimination above and pilot the mechanism against baseline access.

**Expected 2056 Nobel Prize and Turing Award contribution statement (aspirational).** Building on Harsanyi's incomplete-information foundation [3] and Vickrey's incentive-compatible allocation [7], we aspire by 2056 to develop auditable adaptive institutions that join private-information economics, reproducible computation, and behavioral diagnosis. Recognition would require general theory, independent replication, field evidence, equitable safety outcomes, and durable public correction mechanisms; none is claimed achieved here.

#### Open Science Statement

GitHub, Colab 01, Hugging Face, and PS2-FP10-Yiqiao-A0-Poster.pptx point to the same stylized/synthetic model. Checkpoint ffa4836208a3f647181771646c5639286c0726e6 uses Python 3.13.7, NumPy 2.3.5, Matplotlib 3.10.8, nbclient 0.10.2, and Gradio 6.29.0. Shortest reproduction: open Colab 01 and Run all; outputs include BNE, welfare, CCAR, sensitivity, and auction tables/figures. Limits include uncalibrated parameters, two players/types, pure strategies, and no representative behavioral sample.

#### Statement of Contribution to the UN's SDGs

No demonstrated SDG contribution is claimed at proposal stage. Prospective relevance to safe, inclusive urban infrastructure would require a specified target, affected audience, access indicator, safety and confidentiality audit, distributional baseline, field comparison, and a public plan for unintended effects and suspension.

## Author Notes

### Acknowledgements

**Course instructor: Professor Luyao Zhang.** The project uses Python, NumPy, Matplotlib, nbclient, Gradio, and the instructor-provided COMSCI/ECON 206 PS2 Overleaf and poster structures. All computational inputs are stylized/synthetic; no empirical participant dataset was used. OpenAI Codex assisted implementation, debugging, literature-search support, verification, restructuring, and drafting under human review. Appendix D documents Feiyu Li’s dated review of a superseded project version without presenting it as peer validation of the current model. Yiqiao Liu remains responsible for the proposal, citations, computations, and errors.

### Team and individual contributions

**Joint contribution (single-member Team FP10; 126 words).** The integrated contribution is to turn a low-altitude traffic privacy tension into an auditable chain across three disciplines. The project makes commercial-disclosure granularity a Bayesian action, compares equilibrium with a realized-type welfare benchmark, and tests whether reciprocal access to enhanced non-safety services can change incentives without conditioning mandatory safety information. Exhaustive computation exposes both the benchmark improvement and failure regions rather than treating one favorable parameter point as a theorem. A bounded priority-slot comparison separates upstream information sharing from downstream scarcity allocation. The behavioral decision environment then converts the rational prediction, privacy and fairness mechanisms, and a misunderstanding alternative into inspectable Solo and Peer Play prompts. Linked code, notebooks, outputs, and evidence-status labels make the claim reproducible while reserving empirical and policy conclusions for future tests.

**Yiqiao Liu; author order: 1.** Liu decided to separate mandatory safety information from voluntary commercial disclosure and to report CCAR as a candidate, not an optimum. The concrete outputs are the model and tests in `src/` and `tests/`, notebooks in `notebooks/`, behavioral artifacts in `hf_space/` and `hf_static/`, and this paper. Verification comprises exhaustive equilibrium/first-best calculations, seeded simulations, browser checks, hosted Colab Run-all confirmations, and 65 automated passes. The model outputs feed the decision artifact and paper figures. For final submission, Liu is responsible for accurately documenting review scope, rerunning affected checks, finalizing the poster, and approving every claim change.

### References

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- [4] Chinmay Maheshwari, Maria G. Mendoza, Victoria Tuck, Pan-Yang Su, Victor L. Qin, Sanjit A. Seshia, Hamsa Balakrishnan, and Shankar Sastry. 2025. Privacy-Preserving Mechanisms for Coordinating Airspace Usage in Advanced Air Mobility. *ACM Journal on Autonomous Transportation Systems* 2, 4 (2025), 1–34. doi:10.1145/3732290

- [5] Michael Raith. 1996. A General Model of Information Sharing in Oligopoly. *Journal of Economic Theory* 71, 1 (1996), 260–288. doi:10.1006/jeth.1996.0117
- [6] Pan-Yang Su, Chinmay Maheshwari, Victoria Tuck, and Shankar Sastry. 2024. Incentive-Compatible Vertiport Reservation in Advanced Air Mobility: An Auction-Based Approach. arXiv:2403.18166 [econ.TH] doi:10.48550/arXiv.2403.18166
- [7] William Vickrey. 1961. Counterspeculation, Auctions, and Competitive Sealed Tenders. *The Journal of Finance* 16, 1 (1961), 8–37. doi:10.1111/j.1540-6261.1961.tb02789.x

## A Technical details and reproducibility

Nature independently draws  $c_A, c_B \in \{1, 1.6\}$ , each with probability  $1/2$ , and publicly reveals  $d$ . Each operator observes only its own type, then chooses simultaneously. A pure strategy is  $s_i : \{L, H\} \rightarrow \{M, P, F\}$ . Against  $s_j$ , type  $t_i$  evaluates

$$EU_i(a \mid t_i, s_j, d) = \sum_{t_j} \Pr(t_j) u_i(a, s_j(t_j); t_i, d).$$

A profile is a pure BNE iff every prescribed action attains this maximum. The first best separately maximizes  $W = u_A + u_B$  over nine profiles for every  $(c_A, c_B, d)$  and then averages with the prior; it assumes observed types and need not be implementable.

**Table 1: Normalized benchmark inputs.**

| Primitive   | Value                                     |
|-------------|-------------------------------------------|
| Actions     | $M, P, F; x = (0, 1, 2); k = (0, 1, 2.5)$ |
| Types/prior | $c = (1, 1.6); (0.5, 0.5)$ independent    |
| Congestion  | $d = (0.3, 0.6, 1.0)$ public              |
| Technology  | $B = 7.5; \lambda = 0.5$                  |
| CCAR        | $\alpha_M = 0.7$ at Medium/High; others 1 |
| Auction     | 4 bidders; $U[0, 10]$ ; 10,000 rounds     |
| Seeds       | benchmark 206; bid-noise stress 1206      |

### Language-independent pseudocode.

- (1) For every congestion state, enumerate each player’s mappings from types to actions.
- (2) For every strategy pair and own type, average each action’s payoff over rival types; retain pairs in which all prescribed actions are best responses.
- (3) For every realized type pair, enumerate action profiles and retain all welfare maximizers; average equilibrium and first-best welfare under the prior.
- (4) Replace enhanced-access factors according to CCAR, recompute equilibrium utility, and evaluate underlying welfare without counting denied access as a benefit.
- (5) Repeat over confidentiality, Full-cost, access, and congestion grids; record multiplicity and failure diagnostics.
- (6) For the allocation extension, compare first eligible allocation with highest bid/second payment, then repeat seeded draws and bid-noise perturbations.

**Table 2: Interim expected utility of each action against the benchmark equilibrium rival strategy. The rival strategy and chosen action are ordered by own type ( $L, H$ ).**

| Congestion | Own type | Rival strategy | $EU(M)$      | $EU(P)$      | $EU(F)$ | Equilibrium choice |
|------------|----------|----------------|--------------|--------------|---------|--------------------|
| Low        | $L$      | $(M, M)$       | <b>0.000</b> | -0.115       | -1.078  | $M$                |
| Low        | $H$      | $(M, M)$       | <b>0.000</b> | -0.715       | -2.578  | $M$                |
| Medium     | $L$      | $(P, M)$       | 0.885        | <b>1.308</b> | 0.670   | $P$                |
| Medium     | $H$      | $(P, M)$       | <b>0.885</b> | 0.708        | -0.830  | $M$                |
| High       | $L$      | $(P, P)$       | 2.951        | <b>3.741</b> | 3.327   | $P$                |
| High       | $H$      | $(P, P)$       | 2.951        | <b>3.141</b> | 1.827   | $P$                |

**Table 3: Validated equilibrium and welfare outputs.**

| $d$    | BNE ( $L, H$ ) | $W_{BNE}$ | $W_{FB}$ | Gap   |
|--------|----------------|-----------|----------|-------|
| Low    | $(M, M)$       | 0.000     | 0.639    | 0.639 |
| Medium | $(P, M)$       | 2.193     | 3.139    | 0.946 |
| High   | $(P, P)$       | 6.882     | 7.385    | 0.503 |

### A.1 Mechanism sensitivity, actual outputs, and known failures

CCAR scales only the recipient’s enhanced non-safety coordination value: at Medium/High,  $\alpha(M) = \alpha_M$  and  $\alpha(P) = \alpha(F) = 1$ ; at Low all equal one. At  $\alpha_M = 0.7$ , Medium BNE is  $(P, P)$ , underlying welfare is 3.089, and the gap is 0.050. The Medium access sweep gives  $(P, P)$  through 0.70, two asymmetric equilibria at 0.75, and baseline  $(P, M)$  from 0.80. The 760-cell grid counts 401 no-effect, 334 robust-improvement, 18 excessive-disclosure, 3 increased-gap, 66 multiple-BNE, and 0 no-pure-BNE cases; diagnostics overlap and are not probabilities. Lower Full exposure costs can induce Full disclosure. These results show parameter dependence, discontinuous equilibrium-set transitions, and the need for an equilibrium selection rule in multiplicity regions.

### A.2 Reproduction record

Inputs are the stylized primitives above; baseline is unrestricted enhanced access, comparisons are CCAR, FCFS, and truthful second price; metrics are equilibrium actions, welfare/gap, multiplicity, failure flags, revenue, and efficiency. Checkpoint ffa4836208a3f647181771646c5639286c0726e6 was fresh-run with Python 3.13.7, NumPy 2.3.5, Matplotlib 3.10.8, nbclient 0.10.2, and Gradio 6.29.0. The automated suite reports **65 passed, 0 failed**: 27 computational, 6 notebook-portability, 17 Gradio behavioral, and 15 static/export checks. Both hosted Colab 01 and Colab 02 were manually confirmed by the student to complete Run all. Deterministic notebooks are notebooks/01\_strategic\_disclosure\_game.ipynb and notebooks/02\_priority\_slot\_allocation.ipynb; shortest reproduction is open each Colab and Run all. Tracked CSV/PNG/SVG outputs are under outputs/. The behavioral version is the Hugging Face tied to exported provenance data. The validated editable poster is poster/PS2-FP10-Yiqiao-A0-Poster.pptx.

Limitations are two operators, two independent types, three actions, exogenous congestion, pure equilibria, normalized uncalibrated parameters, no implementation-cost model, and no representative behavioral sample. The first best observes types; CCAR

omits enforcement/administration; auction welfare omits urgency, wealth, and interdependent values.

### Open Science Statement

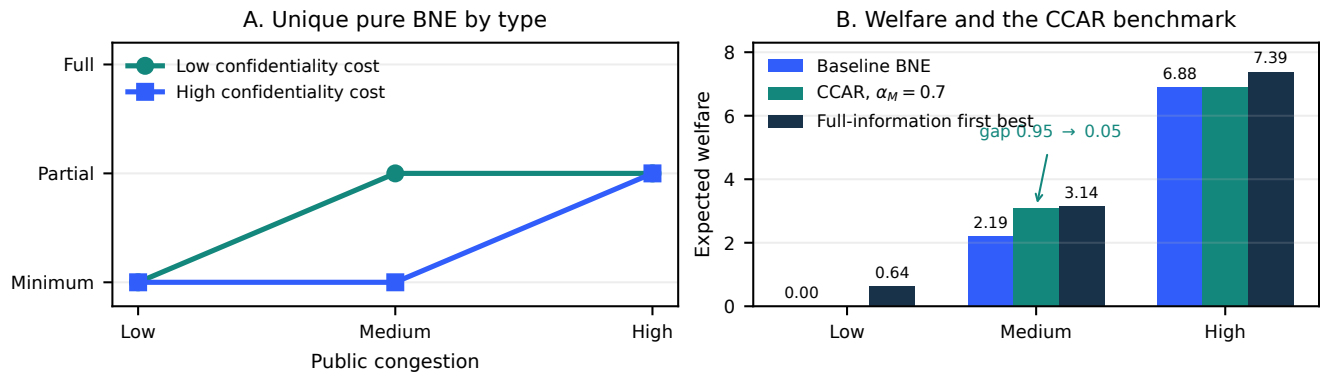
The GitHub, Colab 01, Colab 02, and Hugging Face expose code, stylized/synthetic inputs, dependency files, checkpoint ffa4836208a3f647181771646c5639286c0726e6, and actual outputs. No private dataset, secret, or external LLM is required at runtime. Run-all notebooks reproduce equilibrium, welfare, CCAR, sensitivity, and auction outputs. The repository license and institutional deployment terms govern reuse; absence of empirical calibration and behavioral sampling limits external validity.

### Statement of Contribution to the UN’s SDGs

No demonstrated SDG contribution is claimed at proposal stage. Prospective relevance to safe, inclusive infrastructure requires target-level justification, an access and delay baseline, stakeholder-defined safety/confidentiality outcomes, distributional and unintended-effect audits, a field comparison, and a public correction or suspension rule.

### A.3 AI-use disclosure

**Tool/model and date.** OpenAI Codex, a GPT-5-based coding agent, was used on 2 October 2026. **Purpose.** Assistance covered implementation, debugging, test design, literature-search support, browser and reproducibility checks, document restructuring, and drafting/editing. **Significant prompt categories.** Prompts asked for Bayesian-game enumeration, welfare/CCAR validation, allocation simulation, behavioral-interface construction, deployment, artifact parity, evidence auditing, and official-template compliance. **Accepted suggestions.** Exhaustive-strategy verification, explicit separation of mechanism utility from underlying welfare, bounded auction language, machine-testable evidence labels, and the official A–F reorganization were retained after inspection. **Rejected or qualified suggestions.** Any absolute novelty, universal mechanism superiority, empirical behavioral effect, invented review, invented contact/session metadata, and completed-poster implication were rejected or qualified. **Human revisions.** Yiqiao Liu chose the question, model boundary, parameters, welfare interpretation, CCAR framing, evidence boundary, and final wording. **Verification and responsibility.** Claims were checked against tracked CSVs, code, cited primary sources, 65 tests, notebook runs, browser checks, PDF rendering, and clean-package compilation.



**Figure 2: Tracked computational outputs: type-contingent BNE actions (left) and underlying expected welfare (right). At Medium, CCAR changes  $(P, M)$  to  $(P, P)$  and reduces the gap from 0.946 to 0.050; Low and High coincide with baseline.**

Liu accepts responsibility for every claim, citation, computation, and submission decision.

## B Cumulative Intellectual Development

**Table 4: Auditable development record; topic continuity is not implied.**

| Earlier claim                                                        | Dated evidence/feedback                                                                                                | Revision                                                                                                                                                 | Intellectual consequence                                                                                                                     |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| The earlier “Stay or Switch?” version studied AI memory portability. | Feiyu Li’s 28 Sept. 2026 review raised equilibrium multiplicity and the need for behavioral verification of switching. | The research direction later changed substantially to low-altitude strategic disclosure; the old comments were not relabeled as review of the new model. | The new project retained the methodological demand for robustness, explicit incentives, behavioral evidence boundaries, and reproducibility. |
| A favorable mechanism point could motivate a proposal.               | 2 Oct. 2026 computational checkpoint exposed parameter sensitivity and multiplicity.                                   | Added CCAR sweeps and a 760-cell failure grid.                                                                                                           | The claim became conditional and falsifiable, not universal.                                                                                 |
| The rational model could stand alone.                                | 2 Oct. 2026 behavioral-artifact checkpoint made alternative explanations inspectable.                                  | Added Solo/Peer Play, benchmark reveal, CCAR second choice, and reflection.                                                                              | Future evidence can separate preference/framing from misunderstanding.                                                                       |
| Notebook results were locally reproducible.                          | 2 Oct. 2026 portability checkpoint ffa4836 and manual hosted Run-all confirmation.                                     | Added self-bootstrapping Colab paths and reconciled test counts.                                                                                         | Reproduction now has a public, auditable shortest path.                                                                                      |

## C Open-source and three-lens inspiration

Table 5: Sources and tools actually used.

| Tool/source     | Official URL                                                                                                              | License         | Version/date     | Capability                        | Limitation                    | Design consequence                      |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-----------------------------------|-------------------------------|-----------------------------------------|
| Python          | <a href="https://www.python.org/">https://www.python.org/</a>                                                             | PSF             | 3.13.7           | Reproducible implementation/tests | Environment-dependent         | Locked dependencies and tests           |
| NumPy           | <a href="https://numpy.org/">https://numpy.org/</a>                                                                       | BSD-3-Clause    | 2.3.5            | Arrays and seeded simulation      | Floating-point arithmetic     | Explicit tolerances/seeds               |
| Matplotlib      | <a href="https://matplotlib.org/">https://matplotlib.org/</a>                                                             | PSF-based       | 3.10.8           | Static result figures             | No inference by itself        | Figures read tracked outputs            |
| nbclient        | <a href="https://nbclient.readthedocs.io/">https://nbclient.readthedocs.io/</a>                                           | BSD-3-Clause    | 0.10.2           | Notebook execution                | Hosted runtime differs        | Added portability tests/Colab bootstrap |
| Gradio          | <a href="https://www.gradio.app/">https://www.gradio.app/</a>                                                             | Apache-2.0      | 6.29.0           | Decision-interface prototype      | Not a sampling frame          | Evidence labeled exploratory            |
| acmart          | <a href="https://www.acm.org/publications/proceedings-template">https://www.acm.org/publications/proceedings-template</a> | LPPL            | 2.20, 2026-08-16 | Course paper typesetting          | Layout, not evidence          | Official template organization retained |
| Course template | Instructor-provided repository                                                                                            | Course-provided | Autumn 2026      | Required topology/checklist       | Contains prompts, not results | Five sections and Appendices A–F        |

**Three lenses.** Game theory changes the project by requiring complete type-contingent strategies and interim mutual best responses, predicting  $M/P/F$ . Social choice adds the realized-type first best, welfare gap, disclosure-burden ties, and an explicit efficiency/fairness comparison. Mechanism design changes the feasible institution: CCAR makes enhanced non-safety access reciprocal and demands equilibrium and welfare robustness checks while preserving a mandatory safety floor.

**Week 4 literature stress test.** Chin et al. challenge the congestion/coordination and limited-information dimensions [1, 2]; Maheshwari et al. challenge privacy-preserving coordination [4]; Su et al. and Vickrey challenge scarcity allocation [6, 7]; Raith challenges endogenous strategic information sharing [5]. The surviving gap is only their intersection: type-contingent disclosure granularity with private commercial-confidentiality cost and congestion-dependent coordination value, followed by a separately bounded allocation stage.

Table 6: Closest-literature comparison. Entries summarize substantive overlap; definitions need not be identical.

| Study                        | AAM | Privacy / limited info | Endogenous disclosure | Private conf. cost | Congestion-value link | Auction / allocation | Relationship                                                                                               |
|------------------------------|-----|------------------------|-----------------------|--------------------|-----------------------|----------------------|------------------------------------------------------------------------------------------------------------|
| Chin et al. (2022) [1]       | Yes | No                     | No                    | No                 | Yes                   | No                   | Centralized efficiency/fairness benchmark with dynamic demand.                                             |
| Chin et al. (2023) [2]       | Yes | Yes                    | No                    | No                 | Yes                   | Yes                  | Decentralized protocol minimizes shared trajectory detail and includes optional cost-aware prioritization. |
| Maheshwari et al. (2025) [4] | Yes | Yes                    | No                    | No                 | Yes                   | Yes                  | Privacy-preserving capacity allocation with private vehicle valuations and anonymous prices.               |
| Su et al. (2024) [6]         | Yes | Indirect               | No                    | No                 | Yes                   | Yes                  | Incentive-compatible vertiport reservation with heterogeneous private valuations.                          |
| Raith (1996) [5]             | No  | Strategic info         | Yes                   | No                 | No                    | No                   | General oligopoly incentives to share private demand/cost information.                                     |
| This project                 | Yes | Yes                    | Yes                   | Yes                | Yes                   | Downstream           | Disclosure granularity is a type-contingent action; CCAR and auction remain bounded extensions.            |

## D Review response and revision record

### D.1 Peer review of superseded PS2 version

On 28 September 2026, Feiyu Li (FP3 member 1) reviewed FP10’s earlier AI memory-portability project, “Stay or Switch?” The sheet identifies Session B and Yiqiao Liu. It recognized parameter/model cross-checking and passing-test/reproducibility work; raised the priority concern that equilibrium multiplicity weakened benchmark welfare conclusions; asked whether real users switch when portability exists; suggested participant switching data from the Hugging Face Space; and recommended **Revise**. **This review does not constitute direct peer validation of the final low-altitude drone-disclosure model.**

### D.2 Methodological lessons retained after the research-direction change

The topic changed substantially, so the comments were not treated as direct revisions to the new model. Three lessons carry over: (1) the multiplicity concern reinforced explicit multiple-equilibrium reporting and caution against universalizing one equilibrium; the CCAR grid reports 66 multiple-pure-BNE cells; (2) the request for observed switching reinforced the boundary between computational prediction and behavioral evidence; the Hugging Face artifact is an exploratory classroom exercise, not representative evidence about real drone operators; and (3) the emphasis on checks appears in 65 passed and 0 failed, public GitHub, two runnable Colabs, a public Space, and claim/source audits. These carryovers do not imply endorsement of the current model.

### D.3 Current-version review status

No topic-specific peer review of the final low-altitude drone-disclosure version was available before final submission. The only received peer review concerned the superseded memory-portability version and is therefore used solely as part of the methodological-development record. Earlier-version peer review: **RECEIVED AND DOCUMENTED**. Current-version topic-specific peer review: **NOT AVAILABLE BEFORE FINAL SUBMISSION**. No current-version reviewer question or Keep/Revise/Investigate decision is invented.

## E Structured abstract and cumulative proposal map

Table 7: Official eight-row structured abstract.

| Proposal element                    | Current statement / change                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Background                          | Urban low-altitude operators gain coordination value from flight-intent sharing but may expose commercially sensitive demand; safety disclosure remains mandatory.                                                                                                                                                                                                                                        |
| Motivation and application          | Capacity-constrained drone-delivery corridors require both useful coordination and legitimate allocation of residual commercial priority among stakeholders with different safety, access, and confidentiality concerns.                                                                                                                                                                                  |
| Three connected questions           | Economics asks how congestion and private costs shape BNE versus first best; computation maps welfare and CCAR robustness; behavior tests privacy, reciprocity/fairness, competitive concern, and misunderstanding.                                                                                                                                                                                       |
| Method and model assumptions        | Two-player simultaneous Bayesian game, independent binary types, three disclosure actions, public congestion, exhaustive pure-BNE/first-best enumeration, parameter sweeps, bounded auction, and exploratory decision artifact.                                                                                                                                                                           |
| Results or expected results         | Actual: BNE $(M, M)$ , $(P, M)$ , $(P, P)$ and gaps 0.639, 0.946, 0.503; benchmark CCAR lowers Medium gap to 0.050. Planned: discriminating behavioral and pilot tests; no population effect claimed.                                                                                                                                                                                                     |
| Intellectual merits                 | Integrates endogenous disclosure, auditable welfare comparison, mechanism failure mapping, and behaviorally distinguishable assumptions without claiming absolute novelty.                                                                                                                                                                                                                                |
| Practical impacts                   | Prospective benefit is avoided delay; risks include confidentiality, unequal access, over-disclosure, and misspecification. Adoption needs calibration, safety/distributional review, audit, appeal, and suspension rules.                                                                                                                                                                                |
| Feedback received and revision made | One review concerned a superseded memory-portability version. Because the project later changed direction, it was not treated as direct validation of the current model. Its lessons motivated stronger multiplicity reporting, clearer behavioral-evidence boundaries, and reproducibility checks. No topic-specific review of the final drone-disclosure version was available before final submission. |

**Cumulative proposal map.** The earlier AI memory-portability framing was replaced, rather than cosmetically continued, by low-altitude strategic interaction. Its September 28 review remains scoped to that superseded topic, while its methodological demand for robustness, behavioral verification boundaries, and reproducibility carries forward. The Bayesian model feeds verified computation; computation feeds the Solo/Peer Play benchmark and Figure 2; the bounded auction and real-world governance record state where the analogy and evidence stop. Actual results, planned tests, and the unavailability of current-topic feedback before final submission remain separately labeled.

## F PS2 extension record: auction, linked artifacts, poster, and symposium

### F.1 Auction laboratory record

The scarce resource is one short-window priority slot in a congested corridor. Participants are eligible non-emergency commercial operators; emergency/public-safety flights remain outside payment. Benchmark values/signals are independent private values; reports are sealed bids. In the deterministic case,  $v_A = 8$ ,  $v_B = 5$ ,  $v_C = 3$ ,  $v_D = 2$  and FCFS order is  $B, C, A, D$ . The planned/rational Vickrey bid equals value; truthful bidding is weakly dominant. FCFS allocates to the first eligible request and stops. Second price allocates to the highest bid, breaks ties by ID, charges the second-highest bid, and stops.

Actual computation: FCFS gives  $B$  value 5, efficiency 0.625; truthful second price gives  $A$ , payment/revenue 5, winner utility 3, efficiency 1. Over 10,000 seed-206 rounds, mean FCFS/auction efficiencies are 0.622308/1.000000, allocated values 4.96856/7.99983, and second-price revenue 5.99207. The behavioral stress test uses mean-one lognormal bid noise (seed 1206): efficiency/misallocation is 0.989514/0.1393 at  $\sigma = .1$ , 0.956564/0.2783 at .25, 0.904572/0.4124 at .5, and 0.826281/0.5467 at 1. These are simulations, not classroom observations. Interpretation: truthful IPV allocation is an efficiency benchmark, not a blanket policy ranking; urgency, ability to pay, common values, and institutional legitimacy break the analogy.

### F.2 Linked-artifact parity

| Artifact     | Required parity check                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub       | Same payoff, parameters, BNE, welfare, CCAR, auction outputs, checkpoint ffa4836, runnable paths, and limits.<br>GitHub                                |
| Hugging Face | Same $M/P/F$ setting, benchmark reveal, CCAR, behavioral alternatives, no-sample boundary, and no PII/telemetry.<br>Hugging Face                       |
| A0 poster    | Uses the same questions, results, case, limitations, instructor, acknowledgements, and links. Validated editable file: PS2-FP10-Yiqiao-A0-Poster.pptx. |

### F.3 Symposium and practical-impact record

Symposium session: Session B. The superseded-version review is documented in Appendix D. Current-topic review: **NOT AVAILABLE BEFORE FINAL SUBMISSION**. No current-version presentation question or Keep/Revise/Investigate decision is recorded or invented. The concrete case is commercial drone-delivery operators sharing capacity-constrained urban low-altitude corridors. Stakeholders are operators, the traffic-management authority, customers, nearby/public stakeholders, and emergency/public-safety users. Proposed benefit is coordination and avoided delay. Risks are confidentiality loss, unequal access, over-disclosure, and mis-specified incentives. Missing evidence includes field/pilot calibration, operational latency, confidentiality valuation, safety review, and distributional effects. Correction requires auditable rules, a safety-information floor, appeal/review, and parameter revision or suspension when welfare or access diagnostics fail.